

Pharmacology Lab 1

General Principles of Pharmacology



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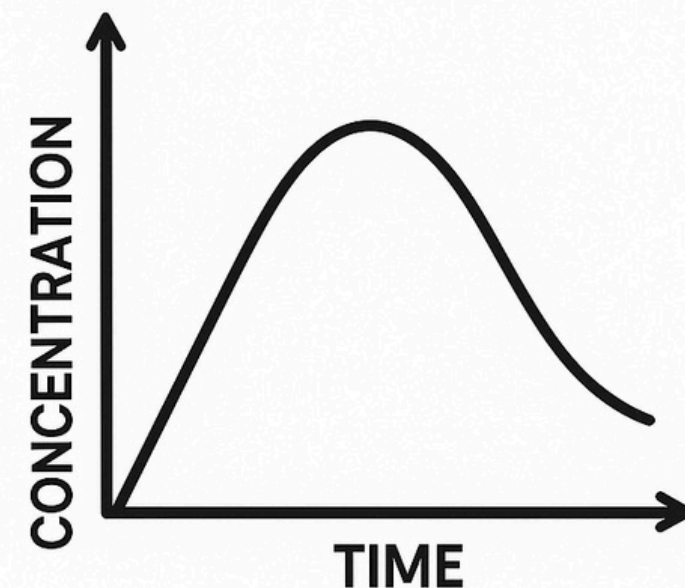
Pharmacology:

is the branch of biomedical science that studies the interactions between drugs and living organisms .

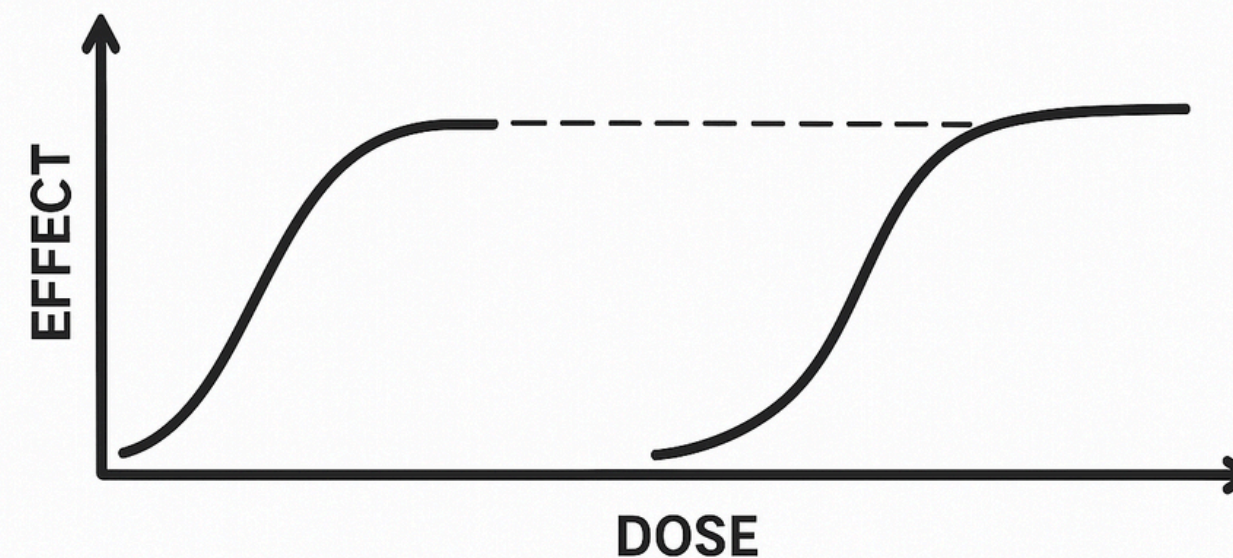


- **Pharmacodynamics** – what the drug does to the body (mechanism of action, effects)
- **Pharmacokinetics** – what the body does to the drug (absorption, distribution, metabolism, elimination)

PHARMACOKINETICS



PHARMACODYNAMICS



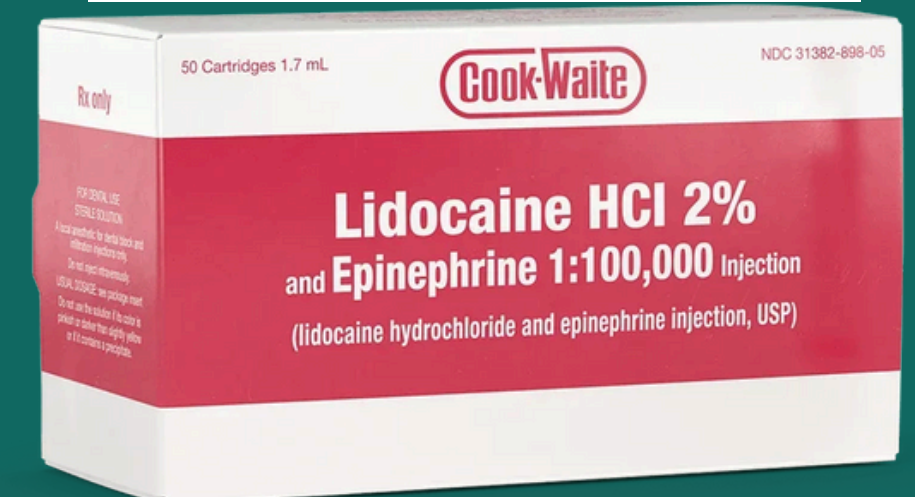
Why Pharmacology Matters in Dentistry?!

1. Pain Management

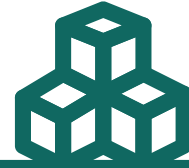
analgesics (e.g., ibuprofen, acetaminophen) and local anesthetics (e.g., lidocaine)

2. Infection Control Understanding antibiotics (e.g., amoxicillin, metronidazole)

3. Anesthesia and Sedation for Safe and effective use of local anesthetics, sedatives, and general anesthetics



4. Management of Systemic Conditions:
(e.g., diabetes, hypertension)



5. Emergency Alertness Dentists must be prepared to manage emergencies like anaphylaxis, hypoglycemia, or cardiac arrest



6. Drug Interactions and Side Effects

To avoid harmful interactions and recognize oral manifestations of systemic drug effects (e.g., gingival hyperplasia from phenytoin)



• Toxicology

Branche of pharmacology studies the harmful effects of chemicals and drugs, including overdose, poisoning, and organ-specific toxicity.



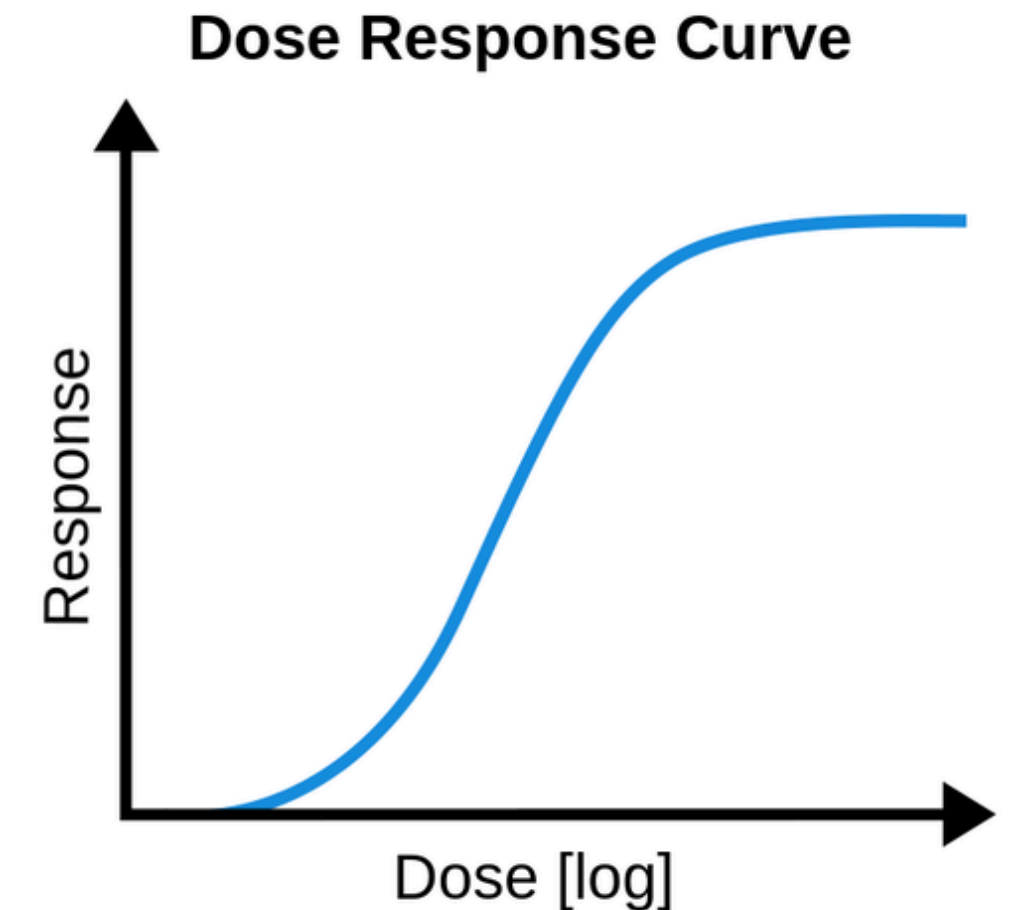
Drugs may be used for:

- Diagnosis (e.g., contrast agents)
- Prevention (e.g., vaccines)
- Treatment or cure of diseases (e.g., antibiotics, antihypertensives)
- Relief of symptoms (e.g., analgesics, antipyretics)



Characteristics of Drugs

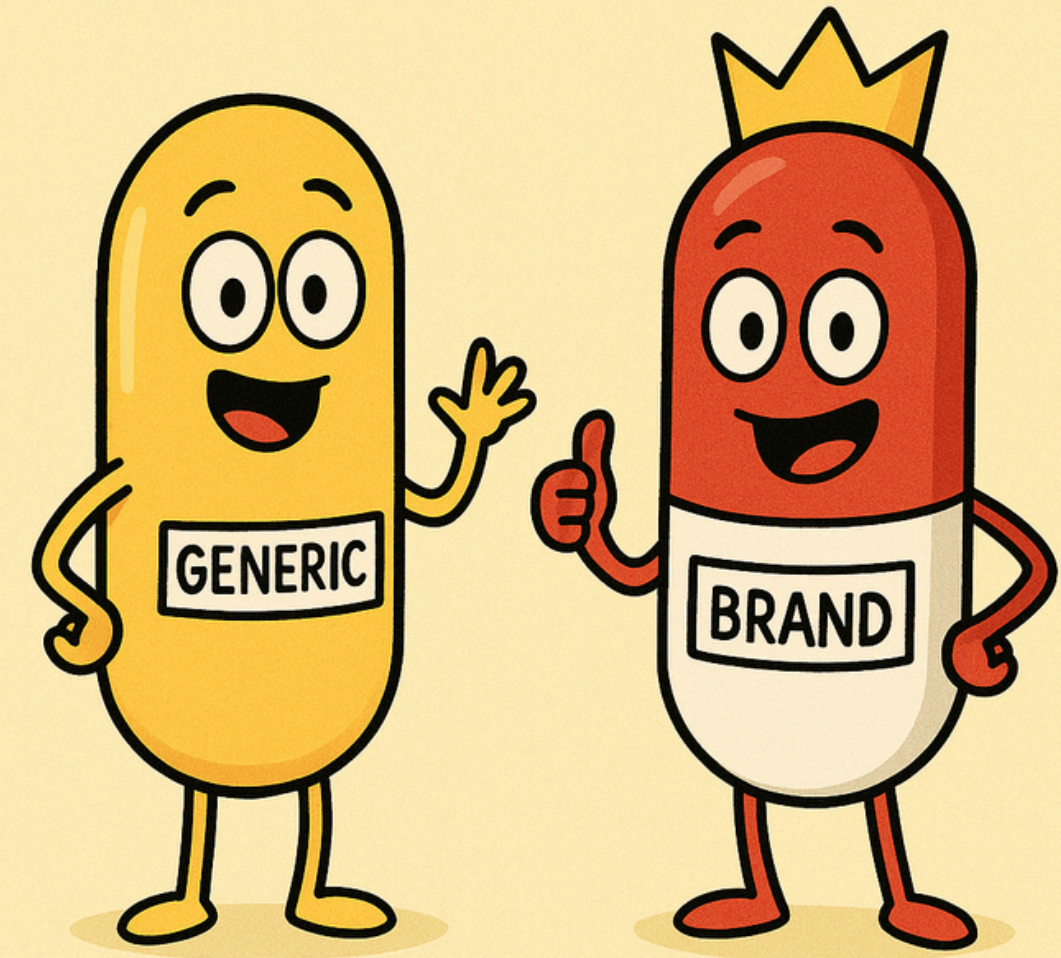
- **Specificity:** Acts on particular cells, tissues, or receptors.
- **Dose-dependent effects:** Therapeutic vs. toxic outcomes
- **Pharmacological activity:** Can be beneficial or harmful.
- **Can be natural, synthetic, or semi-synthetic**



Drug names:

- **Generic name:** Independent of the manufacturer, often indicates the drug group (e.g., amoxicillin).
- **Trade/Brand name:** Given by the manufacturer for marketing purposes (e.g., Amoxil, Trimox).

DRUG NAMES



GENERIC

BRAND

Half-life ($t_{1/2}$)

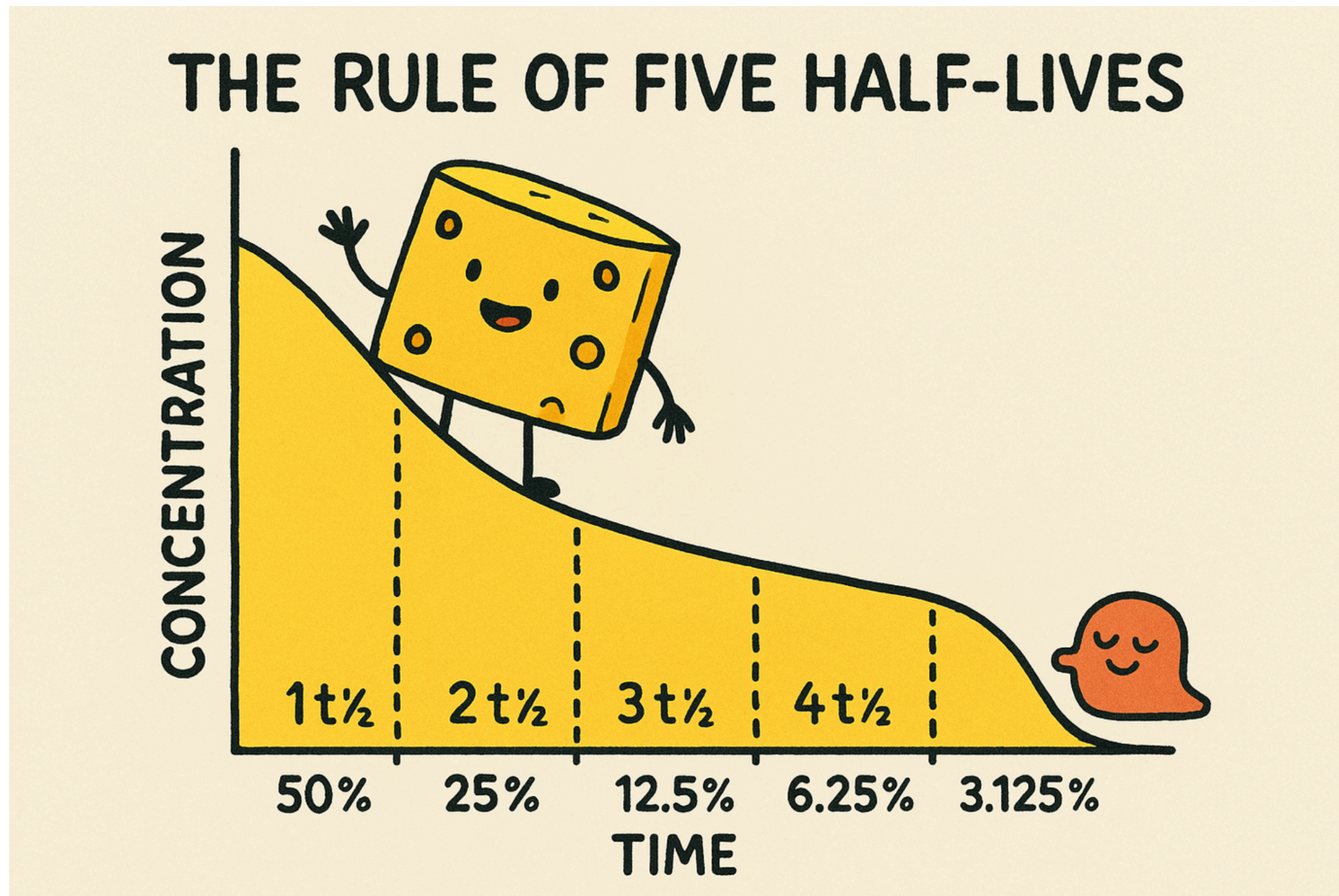
Half-life ($t_{1/2}$) is the time required for the concentration of a drug in the plasma (or body) to decrease by 50% due to metabolism and elimination.

It reflects how long a drug stays active in the body and helps determine:

- **Dosing intervals**
- **Duration of action**
- **Time to reach steady-state**
- **Risk of accumulation or toxicity**

5 half-lives rule :

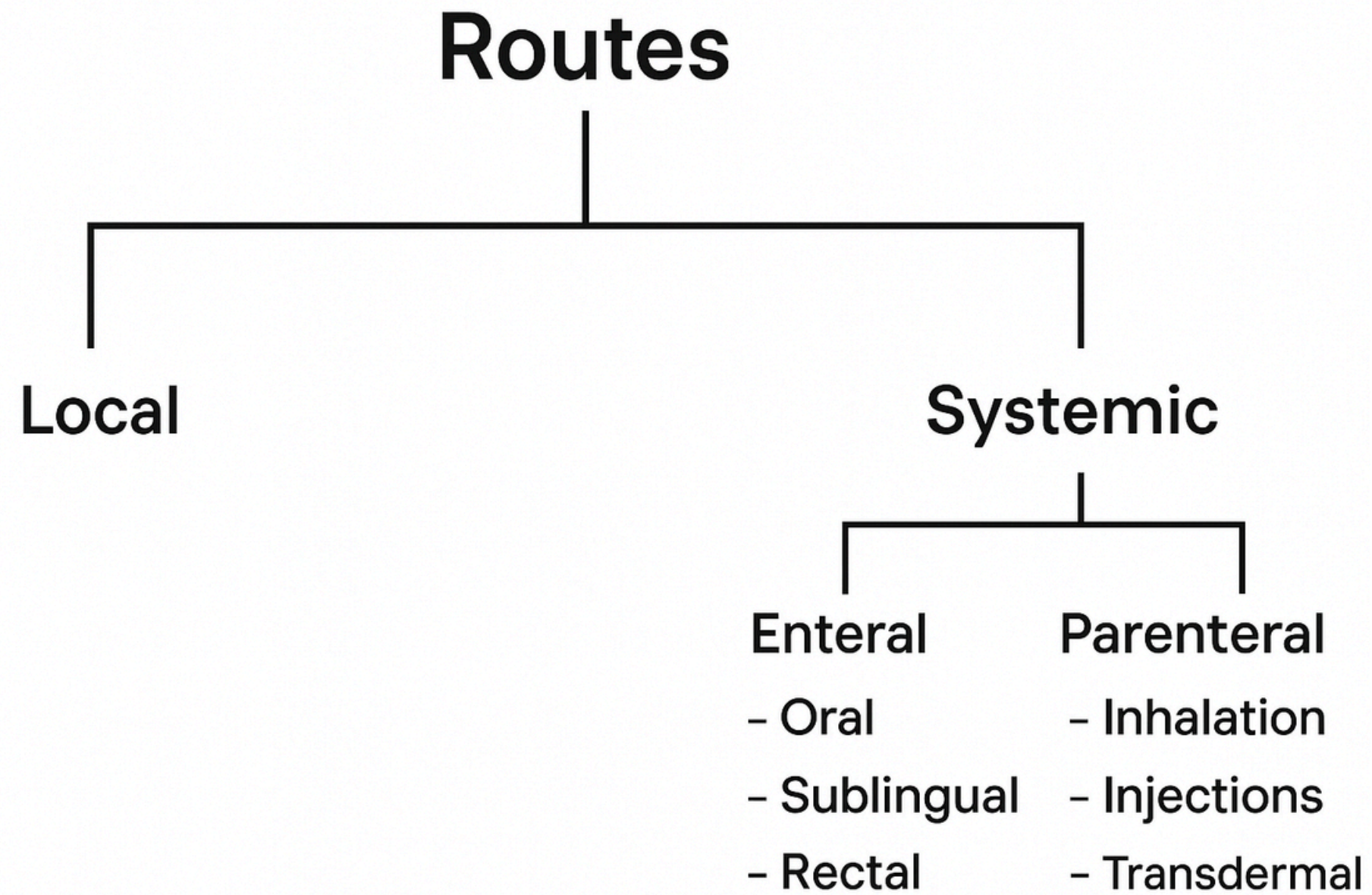
means that the body needs about 5 half-lives to eliminate a drug.



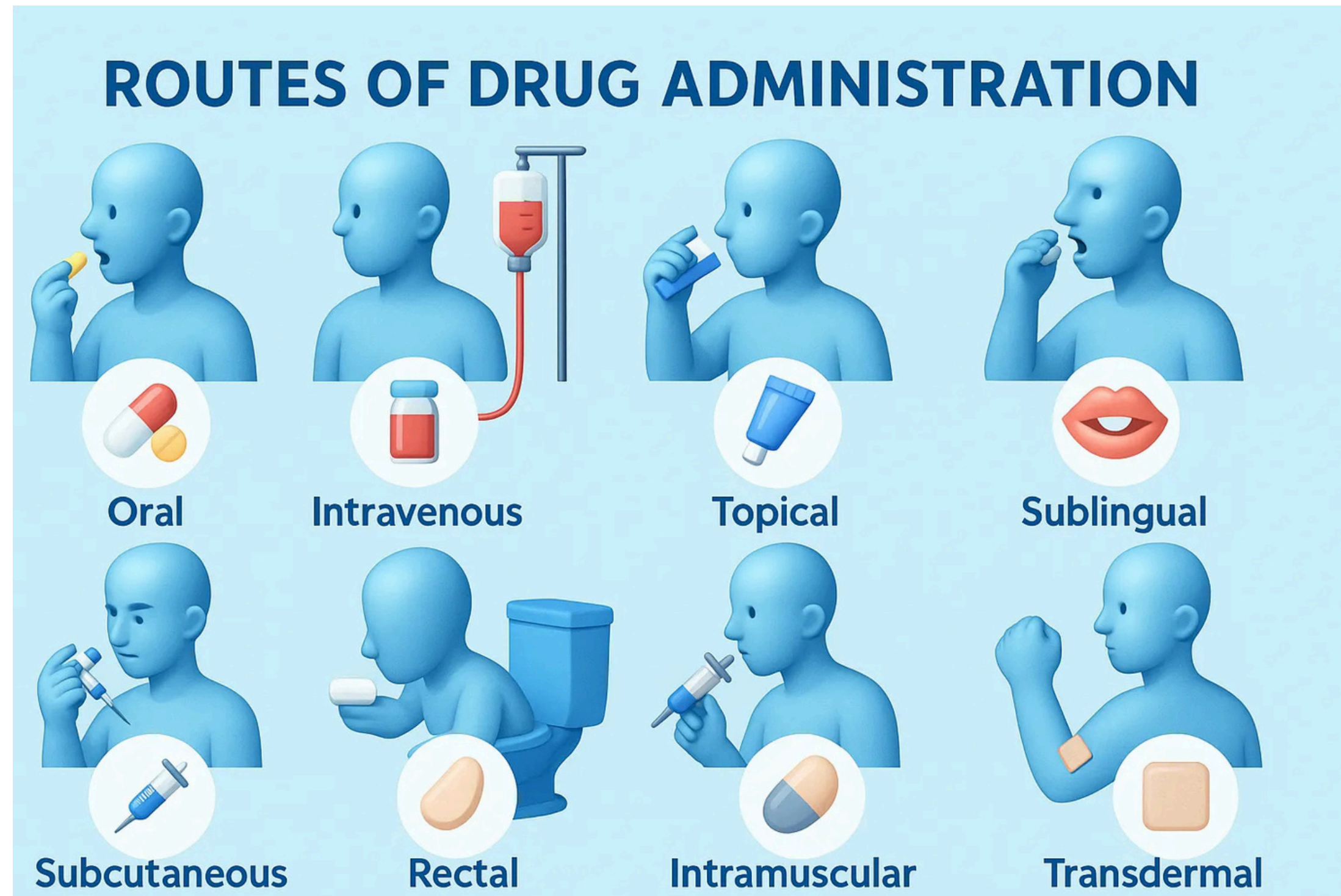
Drugs in Dentistry: Half-life and Clinical Implications

Drug Name	Approximate Half-life	Clinical Implication
Lidocaine	1.5–2 hours	Rapid onset and short duration; ideal for local anesthesia
Amoxicillin	1–1.5 hours	Requires multiple daily doses; effective for dental infections
Metronidazole	~8 hours	Suitable for anaerobic infections; twice-daily dosing
Ibuprofen	2–4 hours	Frequent dosing needed for post-operative pain
Paracetamol	2–3 hours	Safe analgesic; short half-life limits duration

ROUTES OF DRUG ADMINISTRATION:



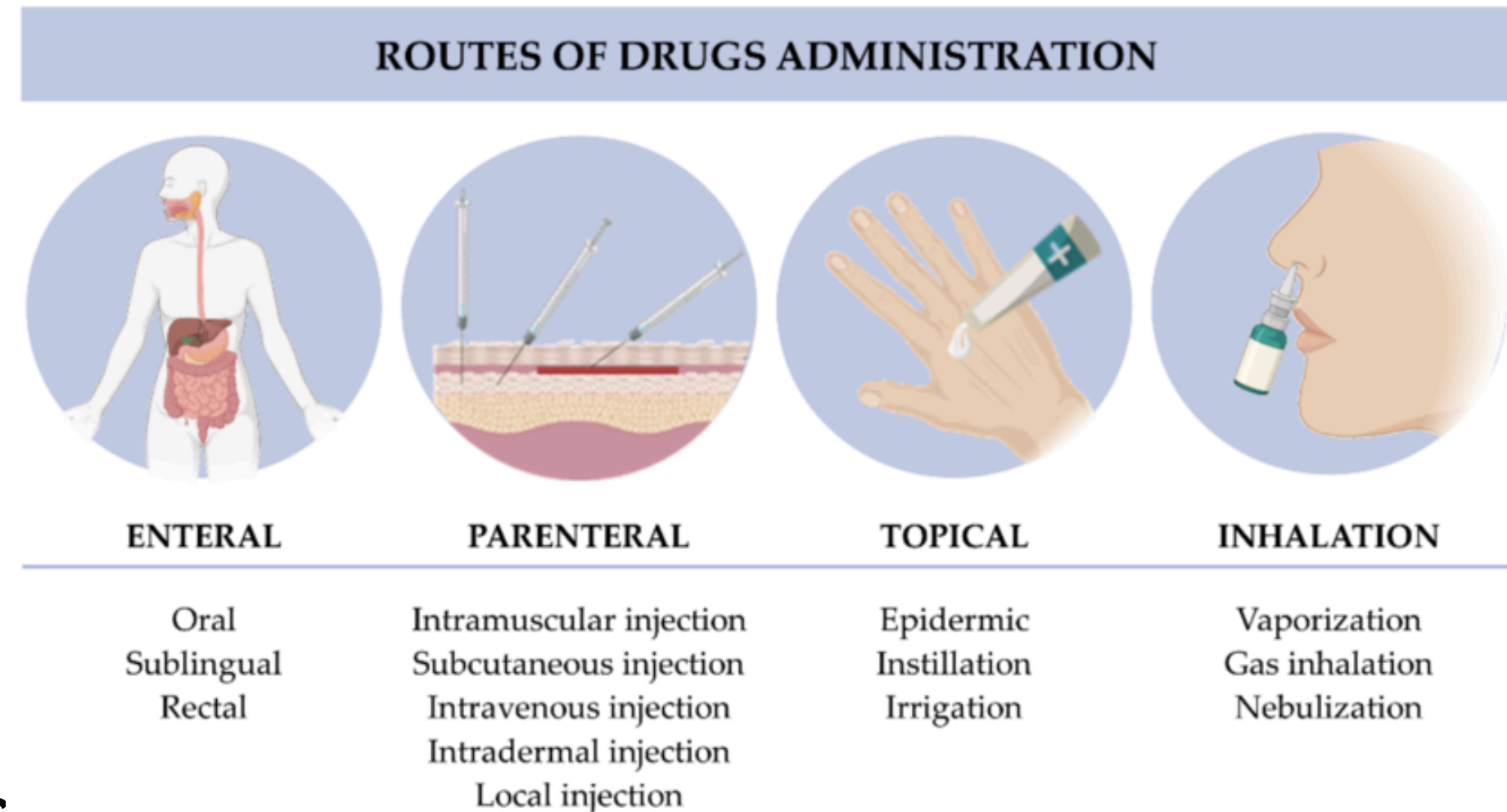
ROUTES OF DRUG ADMINISTRATION:



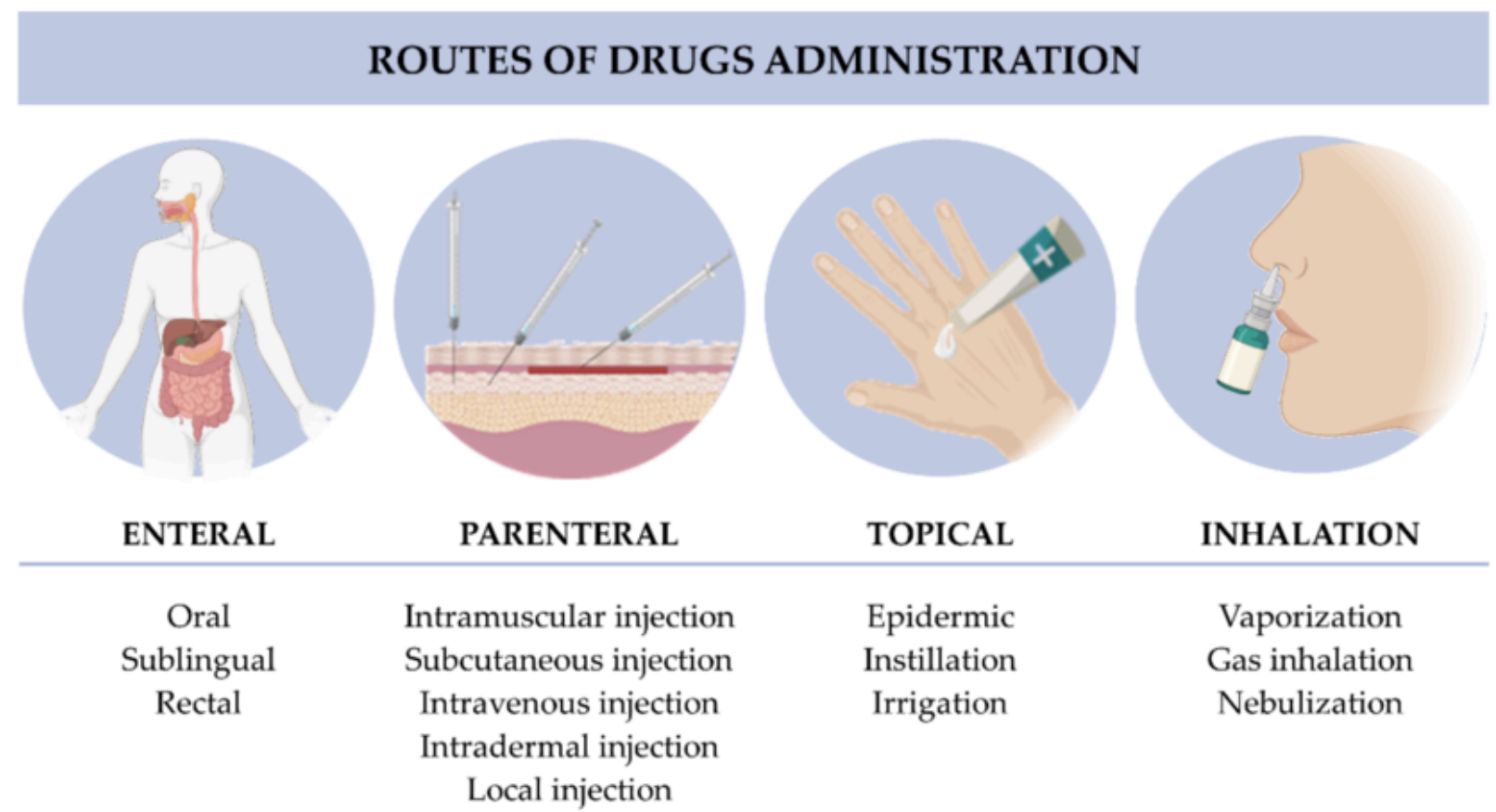
Drug- and patient-related factors determine the selection of routes for drug administration.

The factors are:

1. Characteristics of the drug.
2. Emergency/routine use.
3. Site of action of the drug—local or systemic



- 4. Condition of the patient (unconscious, vomiting, diarrhoea).
- 5. Age of the patient.
- 6. Effect of gastric pH, digestive enzymes and first-pass metabolism.
- 7. Patient's / doctor's choice (sometimes).



Local Routes

It is the simplest mode of administration of a drug at the site where the desired action is required. **Systemic side effects are minimal.**

1. Topical: Drug is applied to the skin or mucous membrane at various sites for local action.

Local Routes of Drug Administration:

- **Oral cavity:** Nystatin (suspension), clotrimazole (troche), acyclovir (cream), lignocaine (ointment/jelly/spray).
- **GI tract:** Non-absorbed tablets, e.g. neomycin (gut sterilization).
- **Rectum/Anal canal:** Suppositories/enemas (local action).



Local Routes

- **Eye, Ear, Nose:** Drops/ointments/sprays, e.g. gentamicin.
- **Bronchi: Inhalations**, e.g. salbutamol, ipratropium (asthma, COPD).
- **Skin:** Ointments, creams, lotions, powders, e.g. clotrimazole (fungal infections).



Local Routes

Other Routes:

- **Intra-arterial:** Rare; angiography, local chemotherapy.
- **Deep tissue injection:** e.g. triamcinolone into joints (rheumatoid arthritis).



Systemic routes

Enteral Routes:

It includes oral, sublingual and rectal routes.

1. Oral Route

It is the most common and acceptable route for drug administration.

Dosage forms are tablet, capsule, syrup, mixture, etc., e.g., paracetamol tablet for fever, omeprazole capsule for peptic ulcer are given orally.



Oral route

Advantages:

- Safer.
- Cheaper.
- Painless.
- Convenient for repeated and prolonged use.
- Can be self-administered.



Oral route

Disadvantages :

Not suitable for emergency as onset of action of orally administered drugs is slow.

It is not suitable for/in:

- Unpalatable and highly irritant drugs.
- Unabsorbable drugs (e.g. aminoglycosides).
- Drugs that are destroyed by digestive juices (e.g. insulin).



Oral route

Disadvantages :

It is not suitable for/in:

- Drugs with extensive first pass metab
- Unconscious patients.
- Uncooperative and unreliable patient
- Patients with severe vomiting and dia



Sublingual route:



Sublingual route:

Details	Sublingual Route
Drug kept under the tongue; absorbed through buccal mucosa → systemic circulation.	Definition
Nitroglycerin (acute angina attack).	Example
- Rapid onset of action - Action stopped by removing/spitting tablet - Avoids first-pass metabolism - Easy self-administration	Advantages
- Not suitable for irritant drugs - Not suitable for lipid-insoluble drugs - Not suitable for drugs with bad smell/taste	Disadvantages

3. Rectal Route

Drugs can be given in the form of solid or liquid.

1. Suppository: It can be used for local (topical) effect as well as systemic effect, e.g. indomethacin for rheumatoid arthritis.

2. Enema: Retention enema can be used for local effect as well as systemic effect. The drug is absorbed through rectal mucous membrane and produces systemic effect, e.g. diazepam for status epilepticus in children.



4-parenteral route:

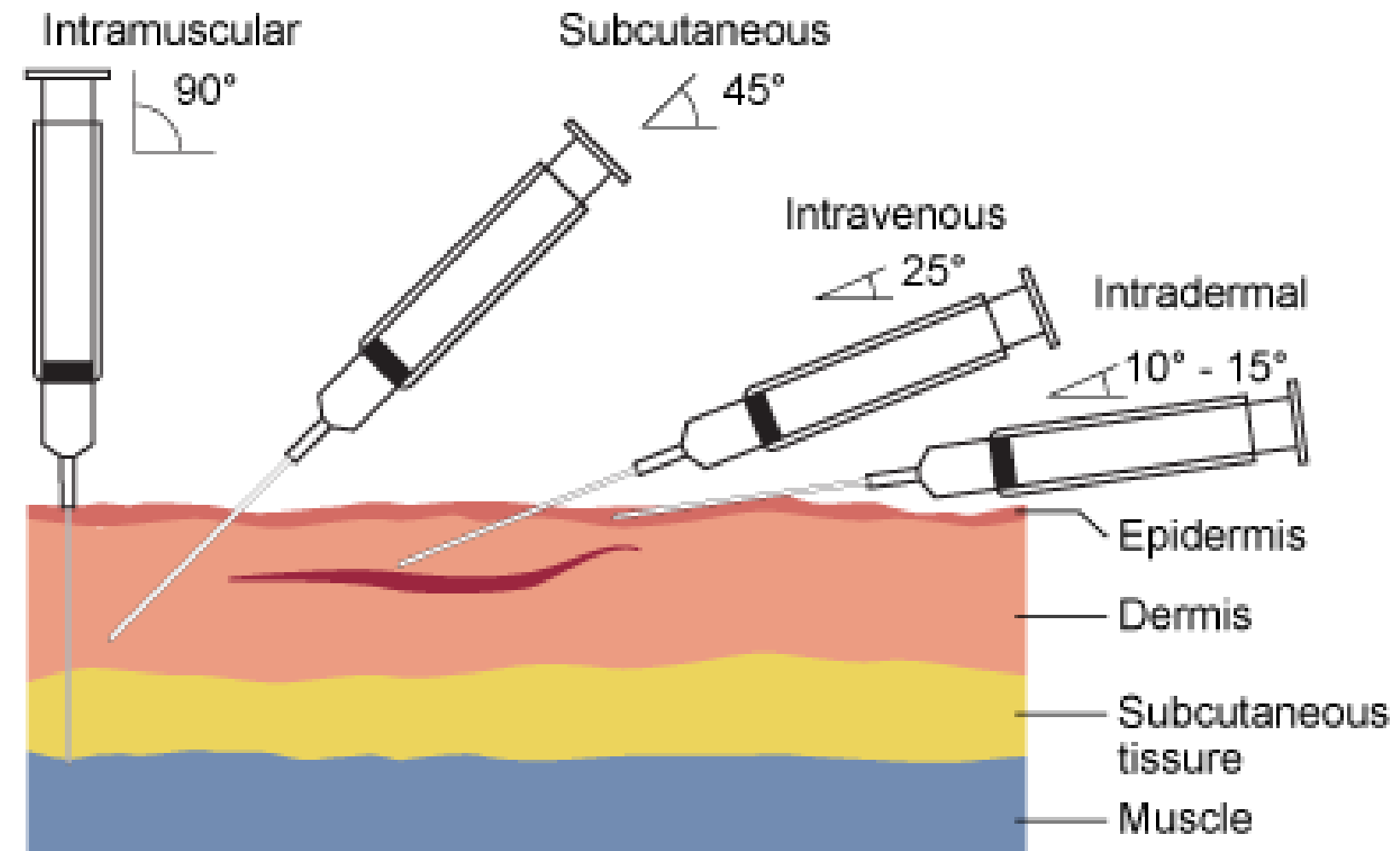
Details	Parenteral Routes
Administration routes other than enteral (non-oral).	Definition
- Fast onset of action (good for emergencies) - Useful in unconscious, uncooperative, or vomiting patients - Suitable for irritant drugs, drugs with high first-pass metabolism, drugs not absorbed orally, or destroyed by digestive juices	Advantages
- Requires aseptic conditions - Preparations must be sterile & expensive - Invasive & painful - Usually not self-administered - May cause local tissue injury (nerves, vessels, etc.)	Disadvantages

5- inhalation route:

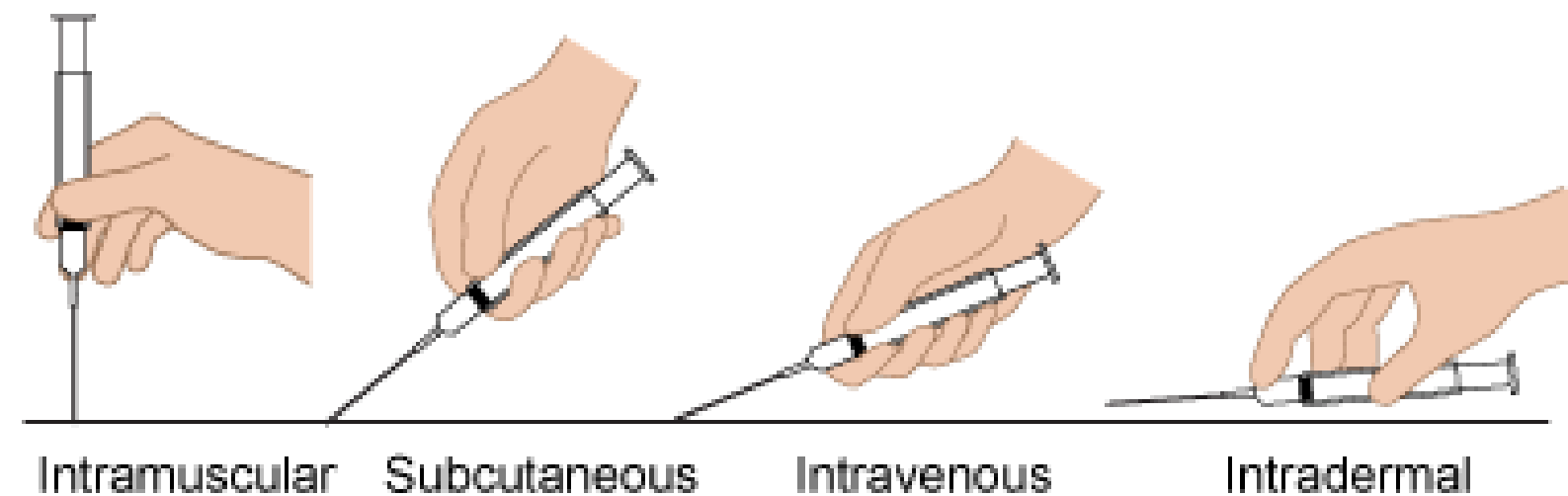
Details	Inhalation Route
Administration of volatile liquids/gases via respiratory tract for systemic effect.	Definition
General anaesthetics.	Example
- Rapid onset of action - Low dose needed → less systemic toxicity - Drug amount can be regulated	Advantages
- May cause local irritation → increased respiratory secretions or bronchospasm	Disadvantages



6- injections:



Angle of injections



6- injections:

Disadvantages	Advantages	Examples	Details	Injection Route
Painful; small dose only	Useful for testing	BCG vaccine, drug sensitivity tests	Into skin layers; only small amount	Intradermal (i.d.)
Only non-irritant drugs; slow absorption; not for emergencies	Self-administration possible; depot drugs can be used	Insulin, adrenaline	Into subcutaneous tissue (thigh, abdomen, arm)	Subcutaneous (s.c.)
Requires aseptic technique; painful; may cause abscess; risk of nerve injury; not self-administered	Faster absorption than oral; mild irritants & depot drugs can be given	Paracetamol, diclofenac	Into large muscles (deltoid, gluteus, vastus lateralis)	Intramuscular (i.m.)



6- injections:

Disadvantages	Advantages	Examples	Details	Route
- Cannot stop action once given - Risk of phlebitis - No self-administration - Requires strict asepsis - Extravasation → tissue damage - Depot preparations not suitable	- 100% bioavailability - Rapid onset (emergency use) - Large volumes can be given - Can give irritant/hypertonic drugs - Maintains constant plasma levels	Diazepam (status epilepticus), IV fluids, dopamine infusion	Direct injection into bloodstream	Intravenous (i.v.)
Requires strict aseptic technique	Direct CNS effect	Spinal anaesthetics (lignocaine), antibiotics (amphotericin B)	Into subarachnoid space	Intrathecal
Risk of cartilage damage with repeated use; strict aseptic precautions	Direct effect on joint; local therapy	Hydrocortisone for rheumatoid arthritis	Into joint space	Intra-articular

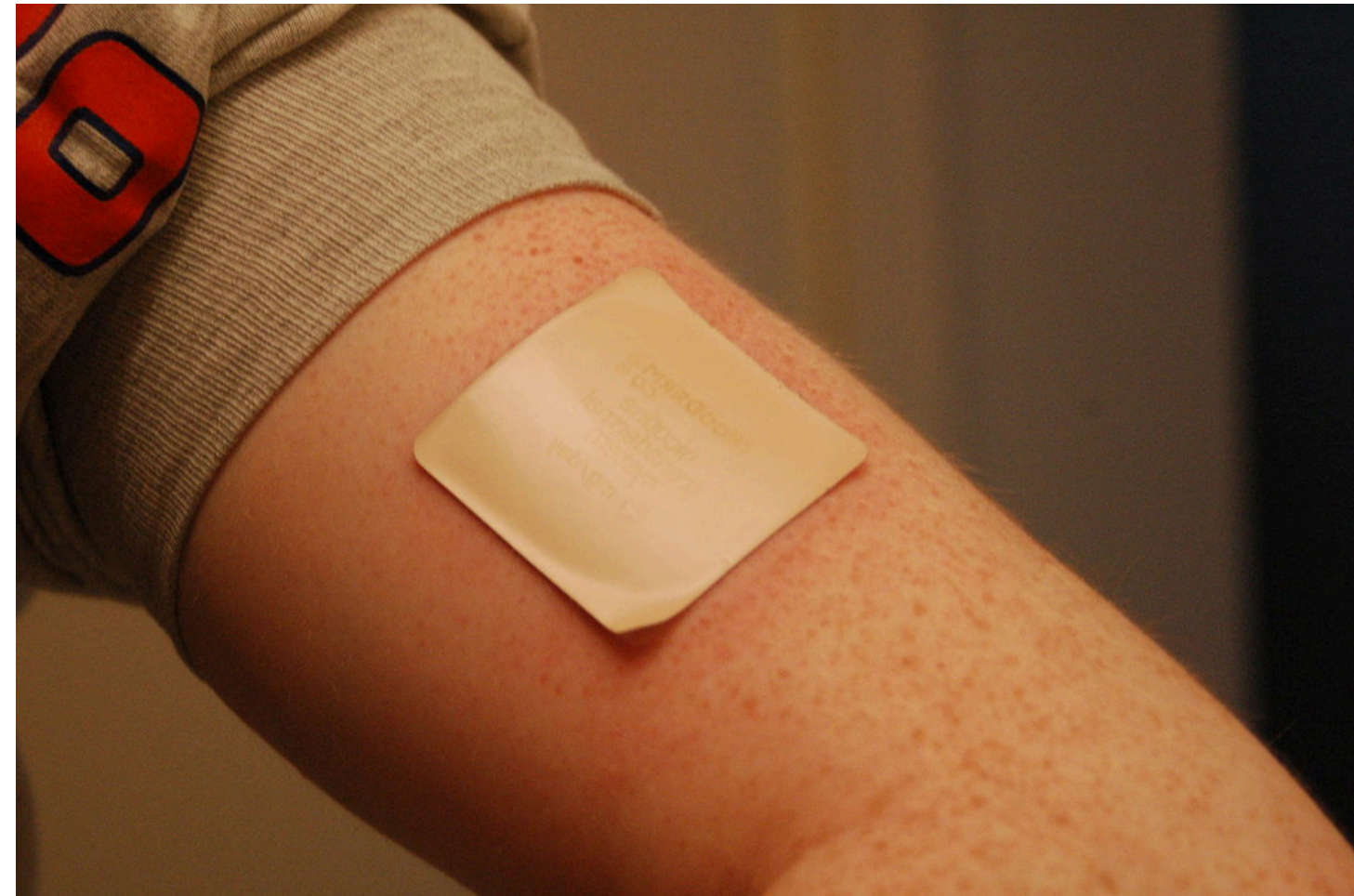


Transdermal Route:

Drug is delivered via patch or ointment for systemic effect, e.g., nitroglycerin patch for angina, oestrogen patch for HRT.

Advantages:

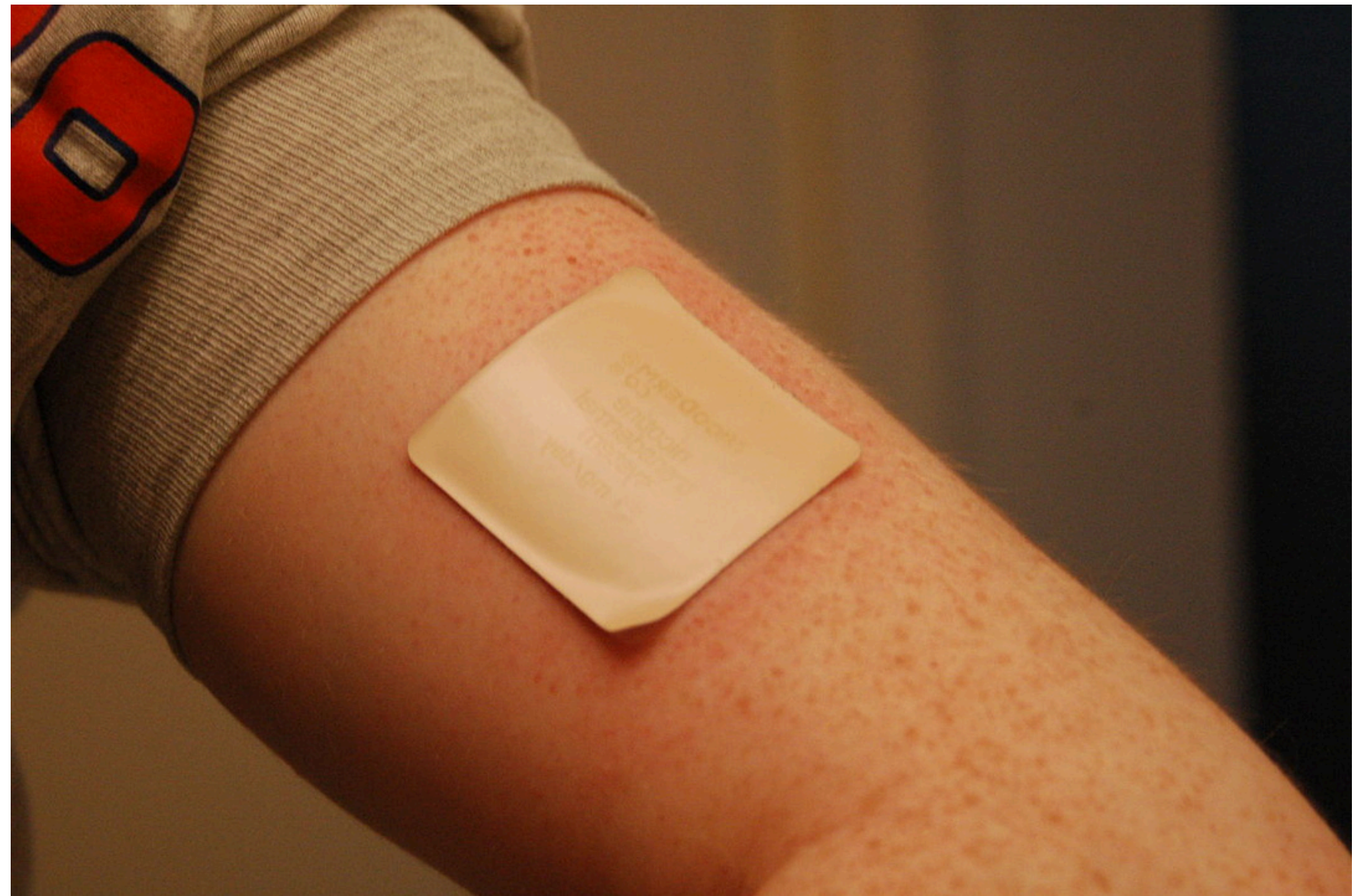
- Can be self-administered.
- Better patient compliance.
- Prolonged duration of action.
- Reduced systemic side effects.
- Maintains constant plasma drug level.



Transdermal Route:

Disadvantages:

- **Expensive.**
- **May cause local irritation (dermatitis, itching).**
- **Patch can fall off unnoticed.**



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Thank you....

روى عن رسول الله (ص):
(من ازداد علماً ولم يزد هدى، لم يزد من الله إلا بعداً)
صدق رسول الله ص